

JOEL POLIZZI

DevOps - ML Ops - AUTOMATION and FRAMEWORK ENGINEER

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San Diego, CA

PROFESSIONAL SUMMARY

- **Infrastructure & Orchestration:** Kubernetes (Deployments, DaemonSets, Services, Ingress), Docker, ContainerD
- **Programming & Scripting:** Python, Bash, CMake, Markup (YAML, TOML), C++, Data Structures
- **Monitoring & Visualization:** Grafana, Custom Dashboard building, COVISE, CalVR, UE4, OpenANTZ, SAGE2
- **Collaboration & Enablement:** Technical documentation, onboarding resources, cross-team training
- **Automation & DevOps:** GitLab CI/CD, GitHub Actions, Jenkins, GitOps, Ansible, PXE Boot
- **Machine Learning Frameworks:** Pytorch Distributed, Horovod, Modin, Ray
- **System Administration:** Linux (Ubuntu, Alpine, Suse, CentOS)
- **Analysis & Reliability Testing:** Static Analysis and regression testing automation

experience

Jul 2022 - Present

Senior Support Engineer Qualcomm

- Architected and maintained GPU-enabled Kubernetes clusters to support scalable machine learning workflows and CI/CD pipelines across Qualcomm AI Research teams.
- Developed internal automation tools such as QGenie for Gitlab Code Reviews, a Flask-based GitLab merge request review agent, reducing manual overhead and improving code quality.
- Designed and deployed infrastructure for GitLab and GitHub Action runners, enabling distributed training and testing of ML models using frameworks like Horovod, DeepSpeed, and PyTorch Distributed.
- Led CI/CD pipeline development and maintenance for key AI projects including AIMET, ML Notebooks, Full Stack Automation, and Nectar, streamlining deployment and reproducibility.
- Collaborated with IT and research teams to optimize cluster health, resource allocation, and deployment strategies, ensuring high availability and performance.
- Provided onboarding, documentation, and training for engineers across Morpheus tracks, improving adoption of DevOps and MLOps best practices.
- Supported lab infrastructure and test device configuration for ML experiments, including automation of static testing and agent deployment workflows.
- Contributed to open-source initiatives and internal documentation to enhance reproducibility and scalability of ML research environments.

Jul 2021 - Jul 2022

Senior Information Technology Engineer Qualcomm

- Linux system administration for the Morpheus team
- Operating system upgrades for 300+ engineers covering workstations and LSF Cluster servers

education

MASTER OF SCIENCE (M.S) DATA SCIENCE (CANDIDATE) Expected Completion in 2027 University of California, San Diego

Relevant Coursework

DSC 210 – Numerical Linear Algebra
DSC 212 – Statistics and Probability
DSC 215 – Statistical Thinking/Experiment Design
DSC 202 – Data Management (RDBS)
DSC 204A – Scalable Data Systems
DSC 240 – Machine Learning
DSC 241 – Statistical Models
DSC 245 – Causal Inference
DSC 260 – Data Ethics and Fairness

BACHELOR OF ARTS (B.A) INTERDISCIPLINARY COMPUTING IN THE ARTS Jun 2014 University of California, San Diego GPA: **3.77** (Cum Laude Honors)

ASSOCIATE OF ARTS (A.A) HUMANITIES/HUMANISTIC STUDIES Jun 2011 Palomar College, San Marcos

SUPPLEMENTAL COURSE WORK (OUTSIDE OF DEGREE PROGRAM) SDCCD (Mesa College, Miramar College)

Relevant Coursework

CISC 190 – Java Programming
CISC 179 – Python Programming
CISC 192 – C/C++ Programming
CISC 211 – Computer Org and Assembly Lang
CISC 187 – Data Structures in C++

	• Collaborate between engineering and IT teams to procure workstations and project specific server configurations • LSF cluster administrator (V100 and A100 nodes)
Nov 2017 – Jul 2021	Research and Development Engineer II UC San Diego, Qualcomm Institute (CalIT2) • Develop state-of-the-art visualization/virtual reality (VR), audio/haptic networked hardware and software. • Cloud native deployment of parallel visualization applications • Devops and support for the Pacific/National Research Platform and CHASE-CI (NSF awards CNS-1456638, CNS-170158, ACI-1540112, ACI-1541349, and OAC-1826967) • Design, recommend, and implement systems and facilities in support of Machine Learning and VR/XR/MR • Programming and database management for visualization platforms (MySQL, PostgreSQL) • Linux system administration (CentOS, Ubuntu 18/20)
Mar 2015 – Nov 2017	AV/IT Systems Technician UC San Diego, Qualcomm Institute (CalIT2) • Maintain and operate Atkinson Hall auditorium, Atkinson Hall Theater, on site Art Gallery, and operate display walls and CAVE systems • Assist, evaluate, and install new technology for the Media Resource Team • Programing for live production (Processing, PureData, Python) • Linux system administration for networked visualization systems
Apr 2014 – Nov 2014	Technology Specialist PSAV • Audio, Video, and IT Specialist
Sep 2013 – Apr 2014	Assistant Recording Engineer UC San Diego • Record musical and theatrical performances for various venues on the UCSD campus. • Studio recording for faculty musicians • Archive recordings
Aug 2006 – Aug 2011	Administrative Clerk Qualcomm • Mail runs, office cleanouts, data entry, stocked breakrooms and supply rooms, occasional reception coverage

key projects and innovations

- SunCave
- Larry Smarr's Future Patient V2
- Covid 19 Report Your Symptoms Visualization Engineering
- Unreal Engine for CAVE systems
- AIMET Simplification and CI/CD refactoring
- Qualcomm ML Apps pipeline and workflow development
- QGenie agent deployment

achievements and highlights

- 2021 CENIC Innovations in Networking Award (team member under Tom Levy)
- National Geographic Channel Drain the Oceans S4. E4
- UCSD Cum Laude Honors, 2014
- Phi Theta Kappa 2009, 2010 Honors

publications

Workflow-Driven Distributed Machine Learning in CHASE-CI: A Cognitive Hardware and Software Ecosystem Community Infrastructure, 2019 Altintas et. al